User-Centric Detection with Interference and Base Station Connection Sparsities

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***Abstract*—In this paper, we investigate the signal detection for user-centric network (UCN) massive multi-input multi-output (mMIMO) system. We consider that the users are divided into multiple user groups (UGs). For each UG, leveraging the interference and base station (BS) connection sparsities, we reveal that the performance of the minimum-mean-square-error (MMSE) detector can be guaranteed by a small number of interfering users and a limited number of associated BSs. Then, we propose a user-centric detection method that decouples the detection process for each UG into two stages. In the first stage, local MF is performed at each associated BSs using local information. In the second stage, group-wise interference cancellation (IC) is carried out to obtain detection results at the primary serving BS (PSBS) of each UG, with information exchanged from auxiliary serving BSs (ASBSs). Simulation results confirm the effectiveness and computational efficiency of our proposed user-centric detection for the UCN mMIMO system.**

***Index Terms*—Base station connection sparsity, interference sparsity, signal detection, user-centric network massive MIMO.**

I. Introduction

Among the wireless technologies, massive multi-input multi-output (mMIMO) has evolved into one of the key technologies in the fifth generation (5G) networks [1], and is expected to maintain its crucial role in the sixth generation (6G) era [2]. By equipping a large number of antennas at the base station (BS), mMIMO can simultaneously serve a number of users in the same time-frequency resource [3], effectively enhancing the spectrum and energy efficiencies [4], [5].

For mMIMO systems, signal detection is of critical significance to harness the potential capacity gains of mMIMO, and has been widely investigated [6]. With large numbers of antennas and users, the maximum *a posteriori* (MAP), which is known as the optimal detection method, suffers from unaffordable computational complexity. Although approaches like sphere decoding [7] and local search [8] can achieve near-optimal performance with decreased complexity, their computational costs remain unacceptable in practice. To facilitate practical application, linear detectors such as zero-forcing (ZF) and minimum mean-squared error (MMSE) detectors have shown progressively significant advantages and garnered much attention due to their appealing low complexity [9].

Nevertheless, cell-edge users in a cellular mMIMO system could suffer from poor quality of service (QoS) due to inherently weak channel conditions and strong inter-cell interference [10]. This issue would become more severe with the further increase in user numbers and the continuous reduction in cell size, necessitating further exploration into systems capable of delivering seamless coverage. To mitigate this limitation, network mMIMO system has been proposed in [11] to enhance the performance of cell-edge users via coherent joint processing [12]. As a result, joint detection in network mMIMO system enables more efficient interference management and improves signal quality at the cell edge [13].

However, as the network size expands and the number of users increases, joint detection of all users by all BSs could lead to unaffordable computational complexity and become impractical. More recently, the user-centric rule has been introduced [14], leading to the user-centric network (UCN) mMIMO system proposed in [15]. In the UCN mMIMO system, the channel gains between each user and most BSs are negligible, indicating that each user can be served by only a limited number of neighboring BSs while maintaining the QoS [16]. This not only eliminates the notion of cell-edge, thereby enhancing the performances of cell-edge users, but also facilitates practical implementation. In addition, the interference between remote users is relatively low, and the detection performance of each user is expected to be determined by a small subset of interfering users in the network. Therefore, this system inherently adopts a distributed architecture, and can effectively reduce the dimension of the observation vector and the complexity of matrix operations for detection, alleviating both computational complexity and implementation challenges.

In this paper, we consider that the users are divided into multiple user groups (UGs), and investigate the uplink (UL) signal detection for UCN mMIMO systems. Leveraging the inherent interference and BS connection sparsities, we reveal that the detection performance of each UG is primarily determined by a small number of interfering users and a subset of BSs. To facilitate practical application, we present a simple user grouping approach and establish effective criteria for

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selecting interfering users and associated BSs for each UG. On this basis, we introduce a user-centric detection method composed of two stages for UCN mMIMO systems. In the first stage, local matched filtering (MF) is carried out at each associated BS for each UG, followed by interference cancellation (IC) at the primary serving BS (PSBS) in the second stage, with necessary information shared from the auxiliary serving BSs (ASBSs).

II. MMSE Detection with Interference Sparsity and BS Connection Sparsity

Consider a network mMIMO system with U single-antenna users served by B BSs, and each BS has N antennas. Let $\mathcal{B} = \{1, 2, \cdots, B\}$ and $\mathcal{U} = \{1, 2, \cdots, U\}$ denote the sets of the BSs and users, respectively. Let $s_i \in \mathbb{C}$ denote the information-bearing signal transmitted by the i-th user with $\mathbb{E}\{s_i s_i^*\} = p_i$ and $\mathbb{E}\{s_i s_j^*\} = 0, j \neq i$, where p_i is the average transmit power of user i. $\mathbf{h}_{i,k} \in \mathbb{C}^{N\times 1}$ denotes the channel vector from the i-th user to the k-th BS.

A. Optimal MMSE Detection

We consider the MMSE detection for the network mMIMO system, where all the users are jointly served by all the BSs. It is assumed that the BSs are synchronized and linked via backhaul links in the network mMIMO system, which enables coherent joint detection.

Let $\mathbf{H}_k = [\mathbf{h}_{1,k}, \mathbf{h}_{2,k}, \cdots, \mathbf{h}_{U,k}] \in \mathbb{C}^{N\times U}$ and $\mathbf{s} = [s_1, s_2, \cdots, s_U]^T$. We assume that the perfect CSI is available at the BS side. The received signal $\mathbf{y}_k$ at BS k can be written as

$$\mathbf{y}_k = \sum_{j\in\mathcal{U}} \mathbf{h}_{j,k} s_j + \mathbf{z}_k = \mathbf{H}_k \mathbf{s} + \mathbf{z}_k, \tag{1}$$

where $\mathbf{z}_k$ is the independent and identically distributed (i.i.d.) circularly symmetric complex Gaussian noise vector distributed as $\mathcal{CN}\left(0, \sigma_z^2 \mathbf{I}_N\right)$. Then, the joint received signal can be expressed as

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \mathbf{z} \tag{2}$$

with $\mathbf{y} = [\mathbf{y}_1^T, \cdots, \mathbf{y}_B^T]^T$ and $\mathbf{H} = [\mathbf{H}_1^T, \cdots, \mathbf{H}_B^T]^T$. Let $\mathbf{\Lambda} \triangleq \mathbb{E}\{\mathbf{s}\mathbf{s}^H\} = \mathrm{diag}\,(p_1, p_2 \cdots, p_U)$. Then, the MMSE detector is given by

$$\mathbf{V}^H = \left(\mathbf{\Lambda}^{\frac{1}{2}} \mathbf{H}^H \mathbf{H} \mathbf{\Lambda}^{\frac{1}{2}} + \sigma_z^2 \mathbf{I}\right)^{-1} \mathbf{\Lambda}^{\frac{1}{2}} \mathbf{H}^H. \tag{3}$$

Among all linear detectors, the MMSE detector is recognized as the optimal detector in the sense of not only minimizing the MSE but also maximizing the instantaneous SINR. Among all detection strategies, joint detection across all BSs is expected to be the optimal approach. Therefore, we refer to the jointly designed MMSE detector (3) as the optimal linear detector for this network mMIMO system.

Generally, we have $BN \geq U$ in the network mMIMO system. Therefore, the design complexity of this optimal MMSE detector is $\left(2BNU^2 + U^3\right)$. As the size of the network expands and the number of antennas at each BS side increases, the overwhelming design and implementation complexities, and the considerable backhaul load may become unaffordable.

B. MMSE Detection with Interference Sparsity

It is worth noting that the optimal MMSE detector $\mathbf{V}^H$ in (3) always does the MF first, and thus $\mathbf{V}^H \in \mathbb{C}^{U\times BN}$ actually lies in the column space of $\mathbf{H}^H \in \mathbb{C}^{U\times BN}$. In this way, when applying $\mathbf{V}^H$ to $\mathbf{y}$, we can define

$$\bar{\mathbf{y}} = \mathbf{H}^H \mathbf{y} = \bar{\mathbf{H}}\mathbf{s} + \bar{\mathbf{z}}, \tag{4a}$$

$$\bar{\mathbf{V}}^H = \left(\mathbf{\Lambda}^{\frac{1}{2}} \mathbf{H}^H \mathbf{H} \mathbf{\Lambda}^{\frac{1}{2}} + \sigma_z^2 \mathbf{I}\right)^{-1} \mathbf{\Lambda}^{\frac{1}{2}}, \tag{4b}$$

where $\bar{\mathbf{y}}$ denotes the matched filtered signal and $\bar{\mathbf{V}} \in \mathbb{C}^{U\times U}$ is the MMSE detector using MF outputs for IC. $\bar{\mathbf{H}} = \mathbf{H}^H\mathbf{H}$ and $\bar{\mathbf{z}} = \mathbf{H}^H\mathbf{z}$ are the channel Gram matrix and the matched filtered noise, respectively. Let $\mathbf{h}_i = \left[\mathbf{h}_{i,1}^T, \mathbf{h}_{i,2}^T, \cdots, \mathbf{h}_{i,B}^T\right]^T \in \mathbb{C}^{BN\times 1}$ denote the stacked channel vector from user i to all the BSs. Then, the matched filtered signal (4a) can be written in a more detailed form as

$$\begin{bmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_U \end{bmatrix} = \begin{bmatrix} \mathbf{h}_1^H\mathbf{h}_1 & \cdots & \mathbf{h}_1^H\mathbf{h}_U \\ \mathbf{h}_2^H\mathbf{h}_1 & \cdots & \mathbf{h}_2^H\mathbf{h}_U \\ \vdots & \ddots & \vdots \\ \mathbf{h}_U^H\mathbf{h}_1 & \cdots & \mathbf{h}_U^H\mathbf{h}_U \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_U \end{bmatrix} + \begin{bmatrix} \bar{z}_1 \\ \bar{z}_2 \\ \vdots \\ \bar{z}_U \end{bmatrix}. \tag{5}$$

Notice that the off-diagonal elements in the channel Gram matrix $\bar{\mathbf{H}}$ can reflect the strength of inter-user interference [17]. Specifically, when $|\left(\bar{\mathbf{H}}\right)_{i,j}| = |\mathbf{h}_i^H\mathbf{h}_j|$ is negligible, it can be seen as there is no inter-user interference between user i and j. For the mMIMO system with a large number of antennas at each BS, users in different directions from the BSs or located far apart inherently satisfy this condition. All these facts suggest that $\bar{\mathbf{H}}$ is numerically sparse, implying that the inter-user interference is limited for each user. We refer to this property as the *interference sparsity*.

For clarity, let $\mathbf{H}_{\mathcal{U}_x,\mathcal{B}_y} \in \mathbb{C}^{B_yN\times U_x}$ denote the channel matrix from $\mathcal{U}_x$ to $\mathcal{B}_y$, and let $\bar{\mathbf{H}}_{\mathcal{U}_x} = \mathbf{H}_{\mathcal{U}_x}^H\mathbf{H}$, where $\mathcal{U}_x$ and $\mathcal{B}_y$ represent arbitrary user and BS sets, respectively, with $\mathrm{card}\,(\mathcal{U}_x) = U_x$ and $\mathrm{card}\,(\mathcal{B}_y) = B_y$. In the following, we focus on the signal detection for $\mathcal{U}_m$ with $i \in \mathcal{U}_m$. Let $\mathcal{I}_m = \mathcal{U}_m \cup \mathcal{U}_m^{\mathcal{I}}$ denote the set of users considered for detecting signals of $\mathcal{U}_m$ with $\mathrm{card}\,(\mathcal{I}_m) = I_m$, where $\mathcal{U}_m^{\mathcal{I}}$ is the set of additional interfering users. We refer to $\mathcal{I}_m = \{i_1, i_2, \cdots, i_{I_m}\}$ as the interference group (IG) of $\mathcal{U}_m$. Let

$$\bar{\mathbf{y}}_{\mathcal{I}_m} = \mathbf{H}_{\mathcal{I}_m}^H \mathbf{y} = \bar{\mathbf{H}}_{\mathcal{I}_m}\mathbf{s} + \mathbf{H}_{\mathcal{I}_m}^H\mathbf{z} \tag{6}$$

denote the partial MF outputs of (4) with $\mathbf{H}_{\mathcal{I}_m}$ and let $\bar{\mathbf{v}}_i\,(\mathcal{I}_m) \in \mathbb{C}^{I_m\times 1}$ denote the designed detector for user $i \in \mathcal{U}_m$ under the signal model specified in (6). With the interference sparsity, as the users not in $\mathcal{I}_m$ cause little interference to $\mathcal{U}_m$, i.e., $\mathbf{H}_{\mathcal{I}_m}^H\mathbf{h}_j \approx 0, j \notin \mathcal{I}_m$, the matched filtered signal (6) for detecting $\mathcal{U}_m$ can be approximated by

$$\bar{\mathbf{y}}_{\mathcal{I}_m} = \mathbf{H}_{\mathcal{I}_m}^H(\mathbf{H}_{\mathcal{I}_m}\mathbf{s}_{\mathcal{I}_m} + \mathbf{z}) + \mathbf{H}_{\mathcal{I}_m}^H \sum_{j\notin\mathcal{I}_m} \mathbf{h}_j s_j \approx \mathbf{H}_{\mathcal{I}_m}^H \tilde{\mathbf{y}}_{\mathcal{I}_m}, \tag{7}$$

where

$$\tilde{\mathbf{y}}_{\mathcal{I}_m} = \mathbf{H}_{\mathcal{I}_m}\mathbf{s}_{\mathcal{I}_m} + \mathbf{z} = \sum_{j\in\mathcal{U}_m} \mathbf{h}_j s_j + \sum_{j\in\mathcal{U}_m^{\mathcal{I}}} \mathbf{h}_j s_j + \mathbf{z}. \tag{8}$$

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(8) can be seen as the corresponding signal model before MF, where the interference outside $\mathcal{I}_m$ is omitted. The MMSE detector for detecting signals of $\mathcal{U}_m$ with (8) is given by

$$\mathbf{V}_{\mathcal{U}_m}^H = \mathbf{\Gamma}_m \left(\mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} \mathbf{H}_{\mathcal{I}_m}^H \mathbf{H}_{\mathcal{I}_m} \mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} + \sigma_z^2 \mathbf{I}_{I_m} \right)^{-1} \mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} \mathbf{H}_{\mathcal{I}_m}^H, \quad (9)$$

where $\mathbf{\Gamma}_m = [\mathbf{I}_{\mathcal{U}_m}, \mathbf{0}_{U_m \times (I_m - U_m)}]$ is a selection matrix. Obviously, $\mathcal{I}_m = U$ leads to the optimal detector (3). However in practice, incorporating all interfering users into each IG can present a considerable complexity challenge. Therefore, it is beneficial to exclude the users with little contribution to the MSE performance from the IG. For clarity, we let $\mathbf{v}_i^\star(\mathcal{I}_m)$ denote the MMSE detector designed for user $i \in \mathcal{U}_m$ with (9), and let $\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m))$ denote the MSE of user i with $\mathbf{v}_i^\star(\mathcal{I}_m)$, which is given by

$$\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m)) = 1 + (\mathbf{v}_i^\star(\mathcal{I}_m))^H \mathbf{R} \mathbf{v}_i^\star(\mathcal{I}_m) - 2\Re\{\sqrt{p_i} \mathbf{h}_i^H \mathbf{v}_i^\star(\mathcal{I}_m)\} \quad (10)$$

with $\mathbf{R} = \mathbf{H}\mathbf{\Lambda}\mathbf{H}^H + \sigma_z^2 \mathbf{I}$. Then, we have the following proposition.

Proposition 1. *Let $\mathcal{U}_{m_2} = \mathcal{U}_{m_1} \cup \{\ell\}$ with $i \in \mathcal{U}_{m_1}, \ell \notin \mathcal{U}_{m_1}$. If*

$$\|\mathbf{h}_j^H \mathbf{h}_\ell\| \le \delta, \forall j \in \mathcal{U}_{m_1}, \quad (11)$$

there exists a constant $c > 0$ such that the following holds

$$\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{U}_{m_1})) - \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{U}_{m_2})) \le c\delta. \quad (12)$$

Further, we have

$$\lim_{\delta \to 0} \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{U}_{m_2})) = \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{U}_{m_1})). \quad (13)$$

The proof is available in [18]. Proposition 1 presents an upper bound of the MSE performance loss of user i when the interference from a particular user is disregarded. According to Proposition 1, omitting the users with small interference with all the users in the UG results in negligible performance loss. As $\bar{\mathbf{H}}$ is inherently a numerically sparse matrix. Leveraging the interference sparsity, Proposition 1 implies that most users in the system just cause minimal interference for each UG, and the detection performance of the MMSE detector (9) can be guaranteed by only using the MF outputs of the intra-group users and a limited number of interfering users.

C. MMSE Detection with Both Interference Sparsity and BS Connection Sparsity

The signal model (8) for detecting signals of $\mathcal{U}_m$ can be written in a more detailed form as

$$\tilde{\mathbf{y}}_{\mathcal{I}_m} = \begin{bmatrix} \mathbf{h}_{i_1,1} & \cdots & \mathbf{h}_{i_{I_m},1} \\ \mathbf{h}_{i_1,2} & \cdots & \mathbf{h}_{i_{I_m},2} \\ \vdots & \ddots & \vdots \\ \mathbf{h}_{i_1,B} & \cdots & \mathbf{h}_{i_{I_m},B} \end{bmatrix} \begin{bmatrix} s_{i_1} \\ s_{i_2} \\ \vdots \\ s_{i_{I_m}} \end{bmatrix} + \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \vdots \\ \mathbf{z}_B \end{bmatrix}. \quad (14)$$

Note that the vectors $\mathbf{h}_{j,k}$ in $\mathbf{H}_{\mathcal{I}_m}$ can reflect the channel strength between users in $\mathcal{I}_m$ and all BSs. To be specific, when $\|\mathbf{h}_{j,k}\|$ is small, it is expected that the received signal of BS k bears little information of s_j. In network mMIMO systems, the magnitude of each element in $\mathbf{h}_{j,k}$ tends to be small for BSs located distant form user j. This suggests that $\mathbf{h}_j$ is inherently a numerically block sparse vector. Therefore, we refer to this property as the *BS connection sparsity*. With this sparsity, the BSs that have strong channel gains with user i constitute the associated BSs of user i denoted by $\bar{\mathcal{B}}_i$. This suggests the possibility of ignoring certain BSs' received signals during detection without significant performance degradation.

Let $\mathcal{B}_m$ denote the set of BSs whose received signals are involved in detecting the signal of the users in $\mathcal{U}_m$ with $\mathrm{card}(\mathcal{B}_m) = B_m$. Specifically, $\mathcal{B}_m$ is termed as the associated BS cluster (BC) of the m-th UG. With the interference and BS connection sparsities, the observation vector of $\mathcal{B}_m$ after MF for detecting $\mathcal{U}_m$ is given by

$$\begin{aligned} \bar{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m} &= \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H \mathbf{y}_{\mathcal{B}_m} \\ &= \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H (\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m} \mathbf{s}_{\mathcal{I}_m} + \mathbf{z}_{\mathcal{B}_m}) + \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H \sum_{j \notin \mathcal{I}_m} \mathbf{h}_{j,\mathcal{B}_m} s_j \quad (15) \\ &\approx \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H (\mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m} \mathbf{s}_{\mathcal{I}_m} + \mathbf{z}_{\mathcal{B}_m}) = \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H \tilde{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m} \end{aligned}$$

with

$$\tilde{\mathbf{y}}_{\mathcal{I}_m,\mathcal{B}_m} = \mathbf{H}_{\mathcal{U}_m,\mathcal{B}_m} \mathbf{s}_{\mathcal{U}_m} + \mathbf{H}_{\mathcal{U}_m^{\mathcal{I}},\mathcal{B}_m} \mathbf{s}_{\mathcal{U}_m^{\mathcal{I}}} + \mathbf{z}_{\mathcal{B}_m}. \quad (16)$$

(16) can be seen as the reduced dimensional signal model for detecting signals of $\mathcal{U}_m$. $\mathbf{z}_{\mathcal{B}_m}$ is the stacked noise. Unlike the signal model in (2), the dimension of the observation vector and the number of interfering users are all reduced in (16). Then, the MMSE detector for detecting signals of $\mathcal{U}_m$ with (16) takes the form of

$$\mathbf{V}_{\mathcal{U}_m,\mathcal{B}_m}^H = \mathbf{\Gamma}_m \left(\mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m} \mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} + \sigma_z^2 \mathbf{I} \right)^{-1} \mathbf{\Lambda}_{\mathcal{I}_m}^{\frac{1}{2}} \mathbf{H}_{\mathcal{I}_m,\mathcal{B}_m}^H, \quad (17)$$

where $\mathbf{\Gamma}_m$ has been defined in (9).

Proposition 2. *Let $\mathcal{B}_{m_2} = \mathcal{B}_{m_1} \cup \{\ell\}$ with $\ell \notin \mathcal{B}_{m_1}$. If $\|\mathbf{H}_{\mathcal{I}_m,\ell}\|_\infty \le \phi$, there exists a constant $p > 0$ such that the following holds*

$$\mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m, \mathcal{B}_{m_1})) - \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m, \mathcal{B}_{m_2})) \le p\phi. \quad (18)$$

Further, we have

$$\lim_{\phi \to 0} \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m, \mathcal{B}_{m_2})) = \mathrm{MSE}(\mathbf{v}_i^\star(\mathcal{I}_m, \mathcal{B}_{m_1})). \quad (19)$$

The proof is available in [18]. Proposition 2 presents an upper bound for the MSE performance loss when the received signal from a certain BS is omitted during the detection process for $\mathcal{U}_m$. From Proposition 2, if the norm of each element in $\mathbf{H}_{\mathcal{I}_m,\ell}$ is negligible, BS ℓ can be excluded from the BC $\mathcal{B}_m$ during the signal detection for $\mathcal{U}_m$ without much performance loss. As discussed previously, $\mathbf{H}_{\mathcal{I}_m}$ is inherently a block sparse matrix, and the condition for (19) is inherently satisfied for the BSs located distant from the IG $\mathcal{I}_m$. With the BS connection sparsity, we reveal that the detection performance of each UG is determined by a small set of BSs in the system.

As the number of interfering users for $\mathcal{U}_m$ is limited, it is typically expected that $U_m \le I_m < B_m N$. In this way, the design and implementation complexities can be reduced to $O\left(\sum_{m=1}^M \left(B_m N I_m^2 + B_m N I_m U_m + I_m^3\right)\right)$ and $\sum_{m=1}^M B_m N U_m$, respectively.

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III. User-Centric Signal Detection

With the inherent interference sparsity and BS connection sparsity, we have demonstrated that the performance of the MMSE detection can be guaranteed by only considering a small number of interfering UTs and a small subset of BSs per UG. However, the detection still involves the cooperation of the BSs in each BC. Generally, a central processing unit (CPU) is required for centralized processing, and all BSs need to exchange information with the CPU, making the system unscalable. To address this issue, we present a user-centric detection approach for UCN mMIMO systems in this section, where the primary served users (PSUs) of each BS constitute a UG and the signal detection of each UG is performed at its PSBS, with necessary information exchanged with the ASBSs.

A. User Grouping and BC/IG Construction for UCN mMIMO System

Before introducing the user-centric MMSE detection, we first clarify how the UCN mMIMO system works. Specifically, we present an efficient approach for user grouping, and provide the construction criteria of the BC and IG for each UG.

In the proposed UCN mMIMO system, each UG experiences interference from a limited number of interfering users and served by a subset of BSs, with one serving as the PSBS and others acting as the ASBSs. Each PSBS is responsible for serving a group of users by locally exchanging the required information with the corresponding ASBSs.

First of all, we present a method for determining the PSBS and associated BSs of each user. In practice, the associated BSs of each user can be identified according to the large-scale fading or the reference signal received power (RSRP). Let

$$\phi_{i,k} = \mathbb{E}\left\{\mathbf{h}_{i,k}^H \mathbf{h}_{i,k}\right\}, \tag{20}$$

which quantifies the mean channel power between user i and BS k. Then, the BS with the best channel conditions for user i can be defined as

$$k_i^\star = \arg\max_{k\in\mathcal{B}} \phi_{i,k}, \tag{21}$$

and is referred to as the PSBS of user i. With Proposition 2, we can define

$$\kappa_{i,\ell} = \phi_{i,k_i^\star} - \phi_{i,\ell} \tag{22}$$

to measure the difference in channel gain experienced by user i from its PSBS relative to BS ℓ. Then, the associated BS set $\bar{\mathcal{B}}_i$ of user i can be constructed by

$$\bar{\mathcal{B}}_i = \left\{\ell \in \mathcal{B} | \kappa_{i,\ell} < \delta_{cbs}^i\right\}, \tag{23}$$

where δ_{cbs}^i is the threshold. Once the associated BSs of each user is determined, the served users of the k-th BS are known and denoted by $\bar{\mathcal{U}}_k$. For clarity, the users whose PSBS is k are termed as the PSUs of BS k and the set is denoted by $\mathcal{U}_k^{\mathrm{P}}$, and the rest users served by BS k are termed as the secondary served users (SSUs), and the user set is denoted by $\mathcal{U}_k^{\mathrm{S}} = \bar{\mathcal{U}}_k \backslash \mathcal{U}_k^{\mathrm{P}}$.

After each user has identified the PSBS, a straightforward and efficient method for user grouping is to let the PSUs of BS k be the k-th UG, i.e., let $\mathcal{U}_k = \mathcal{U}_k^{\mathrm{P}}$. Note that $\mathcal{U}_k$ could be an empty set. As discussed in Section II-B, the users with the same PSBS tend to be geographically closer, and are expected to have relatively strong interference with each other. In this way, the signals for each UG can naturally be processed at their respective PSBSs with necessary information exchanged from ASBSs, which offers improved scalability.

Then, we clarify how to determine the IG for each UG. Let $\tilde{\mathcal{B}}_k = \cup_{i\in\mathcal{U}_k}\bar{\mathcal{B}}_i$ denote the union set of associated BSs associated with the PSUs of BS k. As the BSs in $\tilde{\mathcal{B}}_k$ locate near the users in $\mathcal{U}_k$, it is expected that their served users are more likely to cause relatively strong interference to the users in $\mathcal{U}_k$. Therefore, we propose to select the interfering users from $\cup_{\ell\in\tilde{\mathcal{B}}_k}\mathcal{U}_\ell$ for $\mathcal{U}_k$ rather than from all users. Note that the IG $\mathcal{I}_k$ can be initialized by $\mathcal{U}_k$. Let

$$\psi_{k,j,\ell} = \frac{1}{U_k}\sum_{i\in\mathcal{U}_k}\frac{\|\mathbf{h}_{j,\ell}^H\mathbf{h}_{i,\ell}\|}{\|\mathbf{h}_{i,\ell}^H\mathbf{h}_{i,\ell}\|}, \ell\in\tilde{\mathcal{B}}_k, j\in\mathcal{U}_\ell. \tag{24}$$

If

$$\psi_{k,j,\ell} > \delta_{ig}^k, \tag{25}$$

where $\delta_{ig}^k > 0$, then, j can be included in $\mathcal{I}_k$, otherwise, j is excluded from $\mathcal{I}_k$.

Once the associated BSs of each user and the IG of each UG are determined, the BC of $\mathcal{U}_k$ can be obtained by $\mathcal{B}_k = \bigcup_{i\in\mathcal{I}_k}\bar{\mathcal{B}}_i$. In particular, the BSs in $\mathcal{B}_k\backslash k$ are referred to as the ASBSs of the k-th UG, and the BS set is denoted by $\mathcal{B}_k^{\mathrm{A}}$ with $\mathrm{card}\left(\mathcal{B}_k^{\mathrm{A}}\right) = B_k^{\mathrm{A}}$. In this way, each UG can be served by its PSBS with necessary information exchanged with BSs in $\mathcal{B}_k^{\mathrm{A}}$ in our proposed UCN mMIMO system.

B. Two-Stage User-Centric Signal Detection

In this subsection, we introduce the user-centric signal detection method for UCN mMIMO system. For each UG, the proposed method decouples the detection process into two stages: the first stage is local MF at the baseband unit (BBU) of each associated BS, and the second stage is group-wise IC at the BBU of PSBS with necessary information exchanged from the ASBSs.

We focus on the signal detection for the k-th UG. With the UG, BC and IG determined in Section III-A, the observation vector of $\mathcal{B}_k$ after MF for detecting $\mathcal{U}_k$ is given by

$$\begin{aligned}\bar{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} &= \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\mathbf{y}_{\mathcal{B}_k}\\ &\approx\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H(\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{s}_{\mathcal{I}_k}+\mathbf{z}_{\mathcal{B}_k}) = \mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k}\end{aligned} \tag{26}$$

with

$$\tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} = \mathbf{H}_{\mathcal{U}_k,\mathcal{B}_k}\mathbf{s}_{\mathcal{U}_k} + \mathbf{H}_{\mathcal{U}_k^{\mathcal{I}},\mathcal{B}_k}\mathbf{s}_{\mathcal{U}_k^{\mathcal{I}}} + \mathbf{z}_{\mathcal{B}_k}. \tag{27}$$

$\tilde{\mathbf{y}}_{\mathcal{I}_k,\mathcal{B}_k} \in \mathbb{C}^{B_kN\times 1}$ can be seen as the signal model with interference and BS connection sparsities for detecting signals of $\mathcal{U}_k$. Then, the user-centric detector that minimizes MSE takes the form of

$$\begin{aligned}\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H &= \mathbf{\Gamma}_{\mathcal{U}_k}\left(\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}+\sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\\ &=\mathbf{\Gamma}_{\mathcal{U}_k}\left(\mathbf{G}_{\mathcal{I}_k,k}+\sum_{j\in\mathcal{B}_k^{\mathrm{A}}}\mathbf{G}_{\mathcal{I}_k,j}+\sigma_z^2\mathbf{I}\right)^{-1}\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}\mathbf{H}_{\mathcal{I}_k,\mathcal{B}_k}^H\\ &= \underbrace{\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H}_{\mathrm{MMSE-IC}}\underbrace{\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}}[\mathbf{H}_{\mathcal{I}_k,k_1}^H,\cdots,\mathbf{H}_{\mathcal{I}_k,k_{B_k}}^H]}_{\mathrm{Matched\ Filter}},\end{aligned} \tag{28}$$

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where $\mathbf{\Gamma}_{\mathcal{U}_k} \in \mathbb{C}^{U_k \times I_k}$ is a selection matrix with (i, x_i)-th element being 1 and the other elements being 0, where x_i is the index of user i in $\mathcal{I}_k$. $\mathbf{G}_{\mathcal{I}_k,j} \triangleq \mathbf{\Lambda}_{\mathcal{I}_k}^{1/2} \mathbf{H}_{\mathcal{I}_k,j}^H \mathbf{H}_{\mathcal{I}_k,j} \mathbf{\Lambda}_{\mathcal{I}_k}^{1/2} \in \mathbb{C}^{I_k \times I_k}$ is the channel Gram matrix with transmit power included. As $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ is the MMSE detector for the k-th UG using the MF outputs for IC. Therefore, we refer to $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ as the group-wise MMSE-IC matrix. It can be seen from (28) that the detector $\mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H$ is composed of the group-wise matched filter and the MMSE-IC matrix. With (28), the signal detection for $\mathcal{U}_k$ can be completed by

$$\begin{aligned}\hat{\mathbf{s}}_{\mathcal{U}_k} &= \mathbf{V}_{\mathcal{U}_k,\mathcal{B}_k}^H \mathbf{y}_{\mathcal{B}_k} \\ &= \underbrace{\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H}_{\text{IC}} \sum_{j \in \mathcal{B}_k} \underbrace{\mathbf{\Lambda}_{\mathcal{I}_k}^{\frac{1}{2}} \mathbf{H}_{\mathcal{I}_k,j}^H \mathbf{y}_j}_{\text{Local MF}} = \bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H \sum_{\ell \in \mathcal{B}_k} \bar{\mathbf{y}}_{\mathcal{I}_k,j}, \end{aligned} \quad (29)$$

where $\bar{\mathbf{y}}_{\mathcal{I}_k,j}$ represents the local MF outputs of BS j for detecting $\mathcal{U}_k$. It can be seen that the signal detection for $\mathcal{U}_k$ is completed by local MF followed by group-wise IC with MMSE-IC matrix. Note that the local MF outputs are available to each BS with local CSI and received signal as shown in (29). However, further IC process requires information exchange between BSs. To be specific, the group-wise MMSE-IC matrix computation and signal detection are performed at the PSBS for each UG with information exchanged from ASBSs. For the k-th UG, the group-wise MMSE-IC matrix $\bar{\mathbf{V}}_{\mathcal{U}_k,\mathcal{B}_k}^H$ computation and the signal detection requires $\mathbf{G}_{\mathcal{I}_k,j}$ and $\bar{\mathbf{y}}_{\mathcal{U}_k,j}, j \in \mathcal{B}_k^{\mathrm{A}}$, respectively, as shown in (28) and (29). In this way, the proposed user-centric detection for UCN mMIMO system decouples the detection process into two stages for each UG:

1) **Local MF:** Each associated BS performs MF and computes the channel Gram matrix locally using the channels of intra-group and interfering users.
2) **Group-wise IC:** PSBS computes the group-wise MMSE-IC matrix with the channel Gram matrices exchanged from ASBSs. Then, PSBS applies the MMSE-IC matrix to the sum of local MF outputs for final detection results.

The user-centric nature of our proposed detection is reflected in the individual detection and localized information requirements of each UG. The user-centric detector presented in (28) can be seen as a generic detector with different configurations of IG and BC. Specifically, if $\mathcal{I}_k = \mathcal{U}$ and $\mathcal{B}_k = \mathcal{B}$, (28) becomes the optimal detector (3) for the network mMIMO system. By setting $\mathcal{I}_k = \mathcal{U}_k$ and $\mathcal{B}_k = \{k\}$, (28) is reduced to the MMSE detector for the conventional cellular mMIMO system.

The backhaul load of the k-th BS during the user-centric detection is $\sum_{\ell \in \mathcal{B}_k^{\mathrm{P}}} \left(I_\ell^2 + I_\ell\right)$, where $\mathcal{B}_k^{\mathrm{P}} = \cup_{i \in \mathcal{U}_k^{\mathrm{S}}} k_i^\star$ denotes the set of the PSBSs associated with the SSUs of BS k. The design and implementation complexities of the user-centric detector (28) for all UGs are $O\left(\sum_{k=1}^B \left(B_k N I_k^2 + B_k N I_k U_k + I_k^3\right)\right)$ and $\sum_{k=1}^B B_k N U_k$, respectively.

IV. Simulation Results

We adopt the prevalent QuaDRiGa channel model [19] to generate a simulation scenario, where "3GPP 38.901 RMa NLOS" is considered. To ensure a better coverage, we consider the tri-sector configuration and seven gNodeBs (gNBs) are installed in the system [20]. Each gNB has three co-located BSs and each BS is responsible for a 120-degree coverage [21] as shown in Fig. 1. So there are totally $B = 21$ BSs in the system. The distance between the adjacent gNBs is set to 500 m in our simulations [22]. In the network, the users are distributed in a circle with radius of 750 m. The transmit power of each user is assumed the same and is denoted by p, which is set according to [23]. More detailed system parameters are summarized in Table I. For ease of comparison, we let $\bar{B}_{\mathrm{cbs}}$ denote the number of associated BSs of each user and let I_{ig} denote the number of interfering users of each UG. For brevity, the user-centric detector in (28) that considers I_{ig} interfering users and $\bar{B}_{\mathrm{cbs}}$ associated BSs is termed as UCD with I_{ig} and $\bar{B}_{\mathrm{cbs}}$. The UCD with $I_{\mathrm{ig}} = U$ and $\bar{B}_{\mathrm{cbs}} = B$ is referred to as the optimal MMSE detector, and UCD with $I_{\mathrm{ig}} = U_k$ and $\bar{B}_{\mathrm{cbs}} = 1$ is termed as the MMSE-cellular detector for brevity.

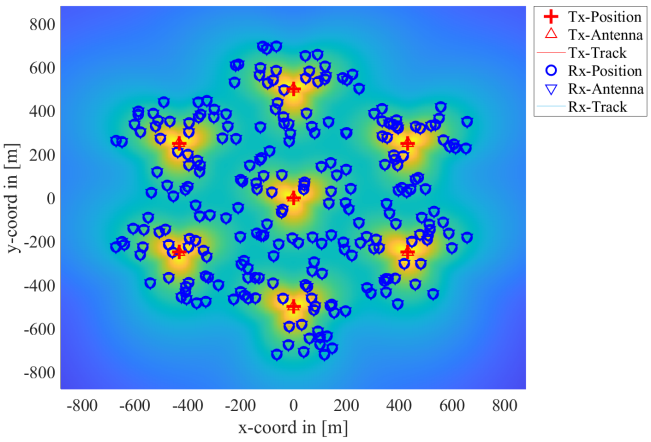


Fig. 1. An illustration of the system layout.

Fig. 2 shows the bit error rate (BER) performance of the user-centric detection for UCN mMIMO system. The UCD with $I_{\mathrm{ig}} = 24$ and $\bar{B}_{\mathrm{cbs}} = 2$ has nearly the same BER performance as the UCD $I_{\mathrm{ig}} = 28$ and $\bar{B}_{\mathrm{cbs}} = 2$. In addition, it has only about 2 dB performance loss compared with $I_{\mathrm{ig}} = U$ and $\bar{B}_{\mathrm{cbs}} = 2$ when $p = 22$ dBm. More importantly, the UCD with $I_{\mathrm{ig}} = 24$ and $\bar{B}_{\mathrm{cbs}} = 2$ has a much better BER performance compared with the MMSE-cellular detector.

Then, we show the design complexity required by the proposed user-centric detection for UCN mMIMO system in Fig. 3. In particular, we let $I_{\mathrm{ig}} = 24$ and $\bar{B}_{\mathrm{cbs}} = 2$ for the user-centric detection, as the UCD under this configuration has shown satisfactory performance according to previous simulation results. From the figure, the design complexity of the UCD is much smaller than that of the optimal MMSE detector, while can achieve comparable performance. To be

TABLE I
Detailed Simulation Parameters

Center frequency	6.725 GHz	U	256
BS antenna type	16×8 UPA	σ_z^2	-112 dBW
Modulation Mode	QAM	Modulation Order	16

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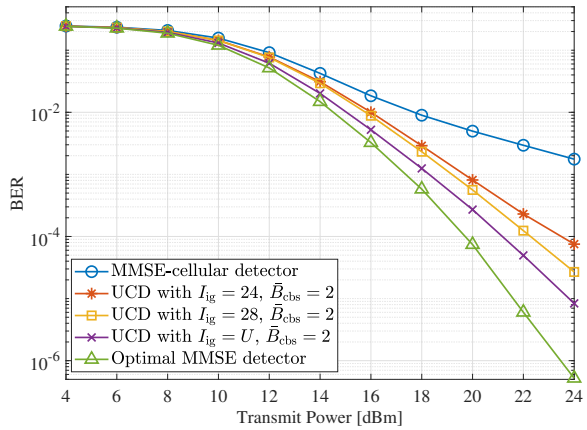


Fig. 2. BER performance of user-centric detector under 16-QAM.

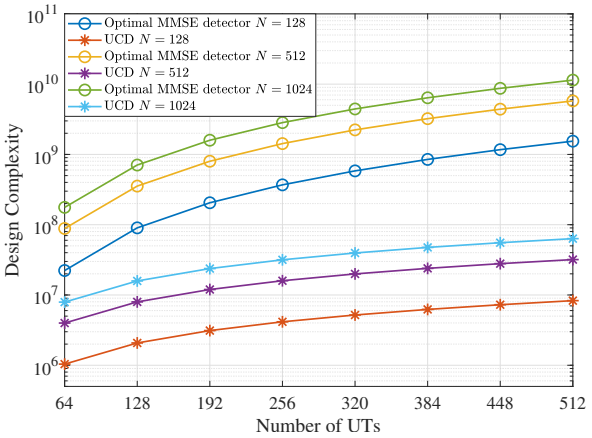


Fig. 3. Comparison of design complexities between the user-centric detector and the optimal MMSE detector.

specific, the design complexity of the UCD is about $\frac{1}{89}$ of the optimal MMSE detector when $N = 128$ and $U = 256$. Moreover, the design complexity of the UCD is only $\frac{1}{180}$ of the optimal MMSE detector when $N = 1024$ and $U = 512$. In fact, as the number of antennas of each BS increases and the size of the system expands, the user-centric detection for the UCN mMIMO system is expected to exhibit even greater advantages.

V. Conclusion

In this paper, we have investigated the signal detection for UCN mMIMO systems. First, with the inherent interference sparsity, we revealed that the detection performance of the MMSE detector can be guaranteed by considering a small number of interfering users. Further, we showed that the detection performance of each UG is determined by a limited number of BSs with the connection BS sparsity. Then, we proposed the user-centric detection approach for the UCN mMIMO system. By first organizing the PSUs of each BS into distinct UGs, we detailed the process for constructing the IG and BC for each UG in UCN mMIMO systems. The user-centric detection operates in a two-stage process. For each UG, local MF is performed at each associated BS using local information in the first stage. In the second stage, group-wise IC is conducted at the PSBS with information exchanged from ASBSs. The simulation results verified the effectiveness and superiority of the proposed user-centric detection for UCN mMIMO systems.

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